

## CHONGQING, China

WMO: 575160

Lat: 29.5758N

Lon: 106.4609E

Elev: 260

StdP: 98.24

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | 2.8        | 3.8 | -2.1                          | 3.3 | 6.3  | -0.2 | 3.8 | 6.1  | 5.8                   | 8.3  | 5.2 | 7.8  | 1.8                   | 0    | 0.335 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 8.1                    | 37.4            | 25.0 | 36.1 | 25.1 | 34.8 | 25.0 | 27.2                | 32.7 | 26.7 | 32.1 | 26.2 | 31.5 | 2.5                  | 70   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 26.0                            | 22.0 | 30.0 | 25.3 | 21.1 | 29.4 | 24.9 | 20.6 | 29.0 | 88.5          | 32.8 | 85.9 | 32.3 | 83.6 | 31.7 | 31.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |  |  |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|--|--|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |  |  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |  |
| 5.8               | 5.0  | 4.3 | 0.5                        | 39.5 | 1.5                | 1.6 | -0.6                                               | 40.7 | -1.4       | 41.6 | -2.3       | 42.5 | -3.3       | 43.7 |  |  |
|                   |      |     | -0.2                       | 28.4 | 1.5                | 0.8 | -1.3                                               | 29.0 | -2.2       | 29.5 | -3.0       | 29.9 | -4.1       | 30.5 |  |  |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 18.8   | 7.8   | 10.2  | 14.4  | 19.3  | 22.7  | 25.3  | 29.1  | 29.1  | 24.5  | 19.0  | 14.4  | 9.2  |
|                                                                               | DBStd       | 7.88   | 2.15  | 2.91  | 3.57  | 3.80  | 3.03  | 2.87  | 2.82  | 3.40  | 3.79  | 3.03  | 2.94  | 2.09 |
|                                                                               | HDD10.0     | 150    | 74    | 29    | 4     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 2     | 41   |
|                                                                               | HDD18.3     | 1152   | 328   | 227   | 129   | 32    | 2     | 0     | 0     | 0     | 1     | 27    | 122   | 284  |
|                                                                               | CDD10.0     | 3361   | 4     | 36    | 142   | 281   | 394   | 459   | 592   | 435   | 279   | 133   | 15    |      |
|                                                                               | CDD18.3     | 1321   | 0     | 0     | 9     | 63    | 137   | 209   | 333   | 333   | 186   | 48    | 3     | 0    |
|                                                                               | CDH23.3     | 13048  | 0     | 0     | 53    | 402   | 919   | 1688  | 4084  | 4117  | 1582  | 200   | 3     | 0    |
|                                                                               | CDH26.7     | 6108   | 0     | 0     | 11    | 117   | 313   | 643   | 2082  | 2208  | 685   | 49    | 0     | 0    |
| Wind                                                                          | WSAvg       | 1.9    | 1.7   | 1.8   | 2.0   | 2.1   | 2.1   | 1.9   | 2.1   | 2.1   | 2.0   | 1.8   | 1.7   | 1.7  |
| Precipitation                                                                 | PrecAvg     | 1101   | 20    | 21    | 37    | 95    | 156   | 168   | 161   | 124   | 144   | 97    | 51    | 26   |
|                                                                               | PrecMax     | 1358   | 49    | 41    | 84    | 194   | 345   | 323   | 424   | 416   | 355   | 193   | 109   | 45   |
|                                                                               | PrecMin     | 789    | 3     | 1     | 12    | 30    | 53    | 18    | 11    | 13    | 32    | 28    | 13    | 1    |
|                                                                               | PrecStd     | 131    | 12    | 10    | 17    | 41    | 69    | 67    | 88    | 83    | 80    | 35    | 23    | 12   |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 15.1  | 20.8  | 28.1  | 32.6  | 34.6  | 36.0  | 38.8  | 39.8  | 38.1  | 30.9  | 23.5  | 15.8 |
|                                                                               |             | MCWB   | 9.6   | 13.9  | 18.9  | 21.3  | 22.3  | 25.7  | 25.3  | 24.6  | 24.3  | 21.5  | 17.4  | 12.3 |
|                                                                               | 2%          | DB     | 13.1  | 17.9  | 24.4  | 29.9  | 32.1  | 34.0  | 37.0  | 38.0  | 35.0  | 27.8  | 21.3  | 14.2 |
|                                                                               |             | MCWB   | 9.3   | 12.5  | 16.8  | 20.3  | 21.7  | 24.8  | 25.5  | 24.7  | 24.4  | 20.4  | 16.4  | 11.5 |
|                                                                               | 5%          | DB     | 11.9  | 16.0  | 22.1  | 27.8  | 30.0  | 32.1  | 35.8  | 36.4  | 32.8  | 25.4  | 19.7  | 13.0 |
|                                                                               |             | MCWB   | 8.8   | 11.8  | 15.7  | 19.5  | 21.4  | 24.2  | 25.6  | 24.9  | 23.7  | 19.8  | 16.0  | 10.5 |
|                                                                               | 10%         | DB     | 10.9  | 14.4  | 20.0  | 25.5  | 28.1  | 30.2  | 34.3  | 35.0  | 30.6  | 23.2  | 18.2  | 12.0 |
|                                                                               |             | MCWB   | 8.4   | 10.9  | 14.7  | 18.8  | 20.9  | 24.0  | 25.5  | 24.9  | 23.1  | 19.1  | 15.5  | 9.8  |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 10.7  | 14.5  | 19.8  | 23.0  | 25.0  | 27.2  | 28.4  | 27.7  | 26.4  | 23.3  | 18.5  | 13.6 |
|                                                                               |             | MCDB   | 13.0  | 19.2  | 26.2  | 30.0  | 30.6  | 32.4  | 33.6  | 33.8  | 33.1  | 28.7  | 21.4  | 14.8 |
|                                                                               | 2%          | WB     | 10.0  | 13.1  | 17.6  | 21.6  | 23.7  | 26.2  | 27.5  | 26.9  | 25.6  | 21.5  | 17.3  | 12.1 |
|                                                                               |             | MCDB   | 12.2  | 16.5  | 22.7  | 27.0  | 28.7  | 31.3  | 32.6  | 33.0  | 31.6  | 25.3  | 20.0  | 13.7 |
|                                                                               | 5%          | WB     | 9.3   | 12.2  | 16.4  | 20.5  | 22.6  | 25.5  | 26.9  | 26.4  | 24.8  | 20.6  | 16.5  | 11.0 |
|                                                                               |             | MCDB   | 11.2  | 15.3  | 20.9  | 25.7  | 27.1  | 30.0  | 32.1  | 32.3  | 30.3  | 23.8  | 19.0  | 12.5 |
|                                                                               | 10%         | WB     | 8.7   | 11.3  | 15.2  | 19.5  | 21.9  | 24.8  | 26.4  | 25.9  | 24.0  | 19.7  | 15.8  | 10.2 |
|                                                                               |             | MCDB   | 10.4  | 14.0  | 19.0  | 24.2  | 26.0  | 28.7  | 31.5  | 31.6  | 29.1  | 22.5  | 17.9  | 11.5 |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 4.5   | 5.5   | 7.0   | 7.7   | 7.6   | 6.9   | 8.1   | 8.4   | 6.8   | 5.1   | 4.8   | 4.2  |
|                                                                               |             | MCDBR  | 6.7   | 8.5   | 11.0  | 11.8  | 11.6  | 10.6  | 10.2  | 10.6  | 10.3  | 9.2   | 7.4   | 5.7  |
|                                                                               |             | MCWBR  | 3.9   | 4.3   | 4.9   | 4.6   | 4.2   | 3.9   | 3.1   | 2.8   | 3.3   | 3.9   | 3.7   | 3.5  |
|                                                                               | 5% WB       | MCDBR  | 5.5   | 7.4   | 9.7   | 9.8   | 8.9   | 8.2   | 8.8   | 9.1   | 8.7   | 7.5   | 6.1   | 4.5  |
|                                                                               |             | MCWBR  | 3.5   | 4.0   | 4.7   | 4.4   | 4.0   | 3.6   | 3.3   | 3.1   | 3.3   | 3.6   | 3.4   | 3.1  |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.783  | 0.814 | 0.834 | 0.806 | 0.731 | 0.697 | 0.656 | 0.693 | 0.719 | 0.831 | 0.855 | 0.836 |      |
|                                                                               | taud        | 1.351  | 1.327 | 1.327 | 1.404 | 1.564 | 1.651 | 1.740 | 1.653 | 1.581 | 1.370 | 1.331 | 1.329 |      |
|                                                                               | Ebn at Noon | 466    | 507   | 542   | 586   | 638   | 657   | 682   | 651   | 609   | 489   | 421   | 407   |      |
|                                                                               | Edh at Noon | 291    | 322   | 341   | 324   | 277   | 253   | 231   | 250   | 260   | 304   | 294   | 286   |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 1.44   | 1.99  | 2.88  | 3.63  | 3.76  | 3.70  | 5.06  | 4.62  | 3.30  | 2.12  | 1.68  | 1.30  |      |
|                                                                               | RadStd      | 0.21   | 0.30  | 0.32  | 0.53  | 0.62  | 0.64  | 0.57  | 0.69  | 0.53  | 0.39  | 0.23  | 0.19  |      |

## Historical Trends

|                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                        |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| Station Only           | N/A   | N/A     | -1.11  | N/A     | -0.41 | -0.52 | N/A         | N/A     | N/A     | N/A     |
| Regional (4 neighbors) | N/A   | N/A     | -0.55  | N/A     | -0.22 | -0.27 | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page